

Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations

Detailed guide and project requirements for the Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations analysis.

Background and Context

In global logistics networks:

- Late deliveries cause SLA breaches, penalties, and customer churn
- Reactive handling (after delays happen) is costly
- Early warnings allow proactive rerouting, reprioritization, or customer communication

While earlier project explains why delays happened, operations teams still need to know: Which orders are likely to be delayed before they are shipped? This Project delivers that forward-looking capability

[Unified Mentor](#)

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Problem Statement

1. APL Logistics faces challenges such as

- Unpredictable shipment delays
- High operational cost of last-minute fixes
- Limited ability to prioritize high-risk orders

2. The organization lacks

- A predictive system to flag late-delivery risk early
- Quantitative risk scores for operational planning
- Explainable insights into delay drivers

Dataset Fields Description

Column Name	Description
Type	Payment method used for the order (e.g., Debit, Credit).
Days for shipping (real)	Actual number of days taken to ship the order.
Days for shipment (scheduled)	Planned or scheduled shipping duration in days.
Benefit per order	Net profit or benefit earned from a single order.
Sales per customer	Total sales value generated by the customer.
Delivery Status	Current delivery status of the order.
Late_delivery_risk	Indicator showing whether the delivery was late (1 = Yes, 0 = No).
Category Id	Unique identifier for the product category.
Category Name	Name of the product category.
Customer City	City where the customer is located.
Customer Country	Country of the customer.
Customer Fname	Customer's first name.
Customer Id	Unique identifier for each customer.
Customer Lname	Customer's last name.
Customer Segment	Segment to which the customer belongs (e.g., Consumer, Corporate)
Customer State	State or region of the customer.
Customer Street	Street address of the customer.
Customer Zipcode	Zip or postal code of the customer.
Department Id	Unique identifier for the department.
Department Name	Name of the department responsible for the product.
Latitude	Latitude coordinate of the customer location.
Longitude	Longitude coordinate of the customer location.

Column Name	Description
Market	Global market region where the order belongs.
Order City	City where the order was delivered.
Order Country	Country where the order was delivered.
Order Customer Id	Customer identifier linked to the order.
Order Item Discount	Discount amount applied to the order item.
Order Item Discount Rate	Discount rate applied to the product price.
Order Item Product Price	Original price of the product before discount.
Order Item Profit Ratio	Profit ratio achieved on the order item.
Order Item Quantity	Number of units ordered for the product.
Sales	Total sales amount generated from the order.
Order Item Total	Final total value of the order item after discount.
Order Profit Per Order	Profit earned from the order after costs.
Order Region	Regional classification of the order.
Order State	State or province where the order was delivered.
Order Status	Current status of the order (e.g., Complete, On Hold).
Product Name	Name of the product ordered.
Product Price	Selling price of the product.
Shipping Mode	Shipping method used for delivery (e.g., Standard, Express).

Data Science Methodology (Step-by-Step)

1.Data Preprocessing

- Handle missing and inconsistent values
- Encode categorical variables (One-Hot / Target Encoding)
- Scale numerical features
- Address class imbalance (SMOTE or class weights)

2.Feature Engineering

Create predictive indicators such as:

- Shipping pressure index (scheduled days vs quantity)
- Mode risk flags (express vs standard)
- Regional congestion indicators
- Order complexity score

3.Model Development

- Baseline Model
Logistic Regression (interpretability benchmark)
- Advanced Models
Random Forest Classifier
Gradient Boosting / XGBoost

4.Model Evaluation

Metric	Business Meaning
ROC-AUC	Overall prediction quality
Precision	Avoid false delay alarms
Recall	Catch as many true delays as possible
F1 Score	Error pattern analysis
Confusion Matrix	Error pattern analysis

5.Risk Probability & Classification

Each order receives:

- Late Delivery Probability (0–1)
- Risk Category:
Low Risk, Medium Risk, High Risk

6.Explainability & Insights

- Feature importance analysis
- Explanation of why an order is high risk
- Transparency for operational trust

Key Performance Indicators (KPIs)

KPI	Description
Late Delivery Probability	Likelihood of delay
Risk Category	Priority for intervention
Key Risk Drivers	Shipping, region, or order factors
High-Risk Order List	Operational action queue

Streamlit Web Application Requirements

Dashboard Modules

1.Delay Risk Overview

- Overall risk distribution
- High-risk order count

2.Order-Level Risk Prediction

- Individual order risk score
- Key contributing factors

3.Region & Mode Risk Analysis

- Risk heatmaps by region
- Shipping mode risk comparison

4.Operations Action Panel

- Orders requiring immediate attention
- Risk-based prioritization

User Capabilities

- Shipping mode filter
- Region / market selector
- Risk threshold slider
- Customer segment filter

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Conclusion

This project transforms APL Logistics' supply chain operations from reactive delay management to predictive risk intelligence. Unlike earlier projects, which analyze past performance and financial outcomes, this project anticipates future delivery failures, enabling early action and smarter logistics decisions. It forms the predictive backbone of an intelligent, data-driven supply chain

[Access Dataset](#)